Haoran Zhang

Software Engineer | Real-Time Rendering and Performance
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Education

University of Southern California — Los Angeles, CA
M.S. in Computer Science — *Expected May 2028*
Coursework: Scientific Computing and Visualization; Analysis of Algorithms

Tsinghua University — Beijing, China
M.S. in Computer Science, GPA: 3.90/4.00 — *June 2026*
Graphics and Geometric Computing Group; Coursework: Photorealistic Rendering, Computer Graphics, Machine Learning

Tsinghua University — Beijing, China
B.Eng. in Architecture, GPA: 3.87/4.00 — *June 2023*

Technical Skills

Languages: C++, Python, GLSL
Graphic Skills: Real-time rendering, rasterization, ray tracing, path tracing, performance optimization
Tools and Libraries: OpenGL, SDL3, CMake, Git, Godot 4, Blender, Open3D

Selected Projects

MiniRenderer | *C++, SDL3, GLSL* — Spring 2025
- Built a CPU rasterization pipeline from scratch with homogeneous clipping, perspective-correct interpolation, depth buffering, texture mapping, and Blinn–Phong shading, without using a rasterization API.
- Achieved over 10× faster ray–scene intersection with a binned surface-area-heuristic BVH; shared camera, geometry, and material abstractions across rasterization and ray-tracing modes.
- Implemented a Monte Carlo path tracer with Cook–Torrance microfacet materials, next-event estimation, and multiple importance sampling.

Collaborative 3D Editor Engine | *C++, Godot 4* — Feb. 2025 – June 2026
- Owned the end-to-end development of a cloud-synchronized file subsystem, spanning cache architecture, asynchronous operations, editor UI, conflict handling, and version recovery.
- Designed a local/remote/baseline cache and hash-based state machine to distinguish one-sided changes from conflicts and provide a consistent source of truth for synchronization decisions.
- Built a serialized background queue for upload, download, delete, copy, and rename workflows, keeping disk and network I/O off the editor's render loop.
- Contributed to custom TCP/RPC networking and operation-based collaboration with thread-safe message queues, binary serialization, version-vector validation, and selective undo/redo.

2D Incompressible Fluid Solver | *C++, SDL3* — Fall 2026
- Implemented a two-dimensional incompressible Euler solver on a staggered MAC grid with semi-Lagrangian advection, finite-difference divergence, iterative pressure solution, velocity projection, and solid-wall boundary conditions.
- Built an interactive SDL3 visualization with a fixed-timestep simulation loop and density/divergence diagnostic views to inspect projection accuracy, numerical behavior, and evolving vortical flow.

Research Experience

Graduate Researcher, Graphics and Geometric Computing Group — Beijing, China
Tsinghua University — *Sept. 2023 – June 2026*
- Developed an interpretable indoor 3D scene-evaluation framework combining quantitative metrics, human-activity simulation, structured scene graphs, and an LLM evaluator fine-tuned on 1,000 annotated scenes.
- Built a training-free pipeline for reconstructing editable large 3D scenes with over 100 entities from a single image, integrating depth estimation, point reconstruction, VLM-based 3D asset generation/alignment, and scene evaluation.

Selected Publications and Award

H. Zhang et al. "The Evaluable Measurements for Interior Digital 3D Scenes." *Journal of Computer-Aided Design and Computer Graphics*, 2025. **Best Paper, CSIAM GDC 2024**.
S.-K. Zhang, L. Yue, **H. Zhang**, et al. "SceneCluster: Interactive Scene Synthesis by Clustering Groups of Furniture Objects." *IEEE TVCG*, 2025.